Smart Agriculture System Based on HarmonyOS

Zhigang Zhang
Faculty of Mechanical Engineering
Xijing University
Xi'an, Shaanxi, China
zhigang1230@qq.com

Heng Chen
Faculty of Mechanical Engineering
Xijing University
Xi'an, Shaanxi, China
chenhenrys@qq.com

Liguo Liu
Chongqing Public Security Bureau
Chongqing, China
185664052@qq.com

Xuli Lu
Faculty of Mechanical Engineering
Xijing University
Xi'an, Shaanxi, China
774313588@qq.com

2023 7th International Conference on Electrical, Mechanical and Computer Engineering (ICEMCE) | 979-8-3503-8287-7/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICEMCE60359.2023.10491072

***Abstract*—In response to the extensive management methods of traditional agriculture, it is impossible to achieve precise and intelligent management of agricultural environmental information. This article is based on the Internet of Things architecture and adopts the HarmonyOS distributed operating system unique to Hi3861. It integrates modules such as SHT30 temperature and humidity sensor, BH1750 light sensor, soil humidity sensor, pump motor, fan, etc. for equipment development in the field of smart agriculture. Through the Huawei Cloud platform, real-time monitoring of the agricultural environment and intelligent management of crop growth status can be achieved, improving agricultural production efficiency and management level.**

Keywords-Smart agriculture; Internet of Things; Hi3861; HarmonyOS

I. Introduction

With the development of society and population growth, traditional agricultural production methods are no longer able to meet the needs of national development and society[1]. The combination of Internet of Things technology and modern agriculture has become an opportunity for the modernization of agriculture, allowing smart agriculture to gradually enter people's lives. Smart agriculture applies Internet of Things technology to traditional agriculture, monitoring agricultural on-site operations through sensor networks and software systems. Therefore, this article designs a HarmonyOS based smart agriculture system for online monitoring and control of crop growth environment, which adds "smart" elements to agricultural production and improves the modernization level of agricultural production[2].

The relevant parameters of crops are processed through the SHT30 temperature and humidity sensor, BH1750 light sensor, and soil humidity sensor through Hi3861, which can display the growth environment of crops in real-time on the Huawei cloud platform. The pump motor can achieve automatic irrigation based on changes in soil moisture, and automatically turn on the fan to cool down when excessive environmental temperature is detected[3]. Users can observe the line chart on the data upload interface for more intuitive analysis. During the development process, the HarmonyOS distributed operating system installed on Hi3861 can ensure the stability, reliability, and scalability of the entire system.

II. Overall System Framework

The Internet of Things cloud platform serves as a transit station for all data. The 2.4GHz Wi Fi built-in in the southbound device Hi3861 is connected to the Huawei cloud platform using the MQTT protocol, while the northbound application web page is connected to the Huawei cloud platform through the HTTPS protocol. We developed a smart agriculture remote control page using Vue3 and Vite, and achieved control and communication of southbound devices by calling the API of Huawei cloud platform. So far, the system architecture from "end" to "management" to "cloud" has been completed[4]. The overall framework diagram of the system is shown in Fig. 1.

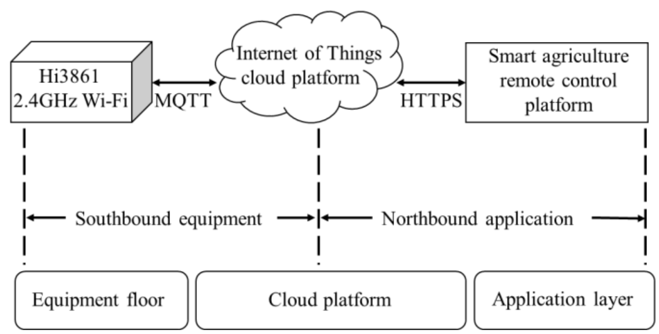

Figure 1. Overall System Framework Diagram

The "end" refers to the terminal composed of the Hi3861 hardware terminal and the Web page, which represent the bottom and top layers of the system, respectively[5]. The Hi3861 is used to collect parameters of the crop growth environment and receive commands from the upper layer.The top Web terminal is used to display the received data and issue instructions to the bottom layer.

"Pipe" is an important component of the Internet of Things, with the Internet of Things cloud platform as the data transmission center. It consists of two parts: one part is the Internet of Things cloud platform and Hi3861 for southbound device data transmission; the other part implements direct data transmission from northbound applications to the Internet of Things cloud platform.

The "cloud" is used to store data. The collected data is uploaded and stored on the IoT cloud platform for instant browsing.

III. System Hardware Design

A. *SHT30 temperature and humidity sensor*

The temperature and humidity sensor model of this system is SHT30 and measures air temperature and humidity through I2C bus commands[6]. SHT30 can provide extremely high reliability and excellent long-term stability, with the advantages of low power consumption, fast response, strong anti-interference ability, and high accuracy. The SHT30 temperature and humidity sensor has a power supply of 3.3V and a measurement humidity range of 0% to 100%. The SHT30 circuit is shown in Fig. 2.

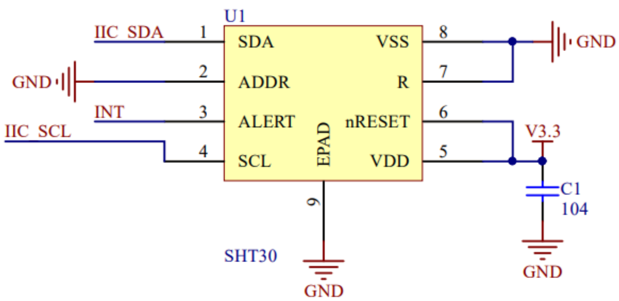

Figure 2. SHT30 Circuit

B. *BH1750 Light Sensor*

The light sensor of this system uses BH1750, which includes a photosensitive resistor in the sensor part. When light shines on the photosensitive resistor, the resistance value of the resistor will change. BH1750 converts this change into a voltage signal through a circuit and performs ADC conversion to obtain a digital value representing the current light intensity[7]. The data is then processed and stored through I2C. The circuit design of the BH1750 light sensor is shown in Fig. 3.

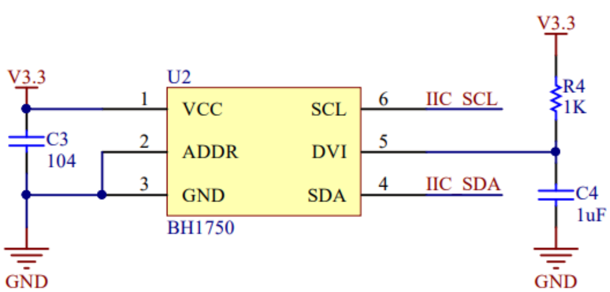

Figure 3. BH1750 Circuit

C. *Water pump drive circuit*

Use optocoupler TLP521 and transistor to form the water pump driving circuit. When the humidity of the soil is detected to be less than the set threshold, Hi3861 sends a signal to turn on TLP521, the transistor S8050 is turned on, and the water pump works to water the vegetation or crops[8]; When the soil humidity reaches the set threshold, the water pump automatically shuts down. The water pump drive circuit is shown in Figure 4.

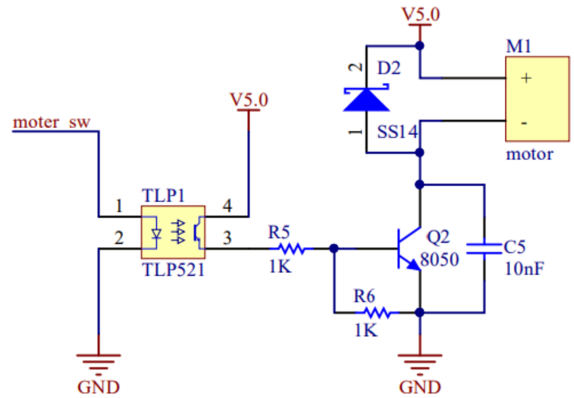

Figure 4. Water Pump Drive Circuit

IV. Design of Embedded System Based on HarmonyOS

A. *The HarmonyOS development environment based on W-Fi IoT is shown in Fig. 5.*

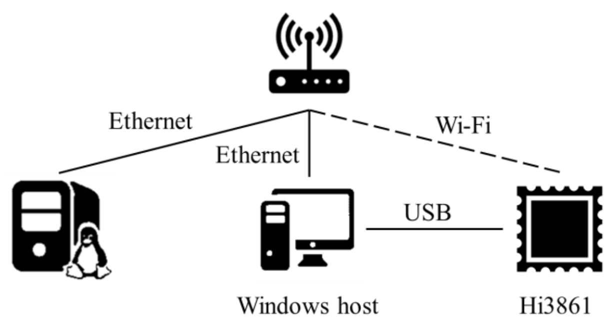

Figure 5. HarmonyOS Development Environment

This system uses the Ubuntu 18.04 system as the Linux compilation server, and maps the obtained HarmonyOS system source code to the Windows platform for code editing. After code compilation on Ubuntu, generate a burnable binary file and burn it to the development board. Finally, the MobaXterm integrated serial port tool can be used to print and output logs.

B. *Data transmission implementation*

This system creates access products on the Huawei IoT cloud platform, and generates corresponding cloud platform IP, port number, and Device after adding devices ID, paired with the key. After adding a custom model to the device, add the attribute names, data types, and related control commands for controlling the pump motor, fan, and other uploaded data. The uploaded data will be displayed in real-time in the configured product. In addition, only by ensuring that the device and cloud are within the same local area network can there be correct upload actions, so WIFI connection judgment should be performed first. The entire software flowchart is shown in Fig.6.

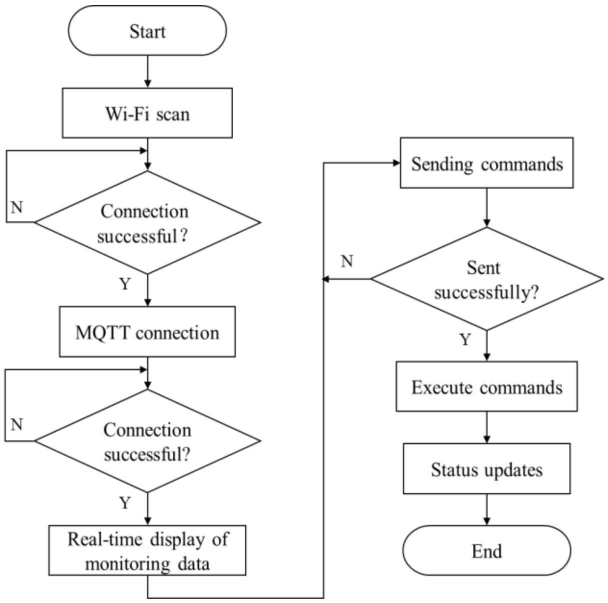

Figure 6. Software Development Flowchart

C. *Huawei Cloud Platform Application Development*

Huawei Cloud is Huawei's cloud service brand, which opens Huawei's more than 30 years of technology accumulation and product solutions in the ICT field to customers, providing stable, reliable, secure, and sustainable cloud services.The development of the Huawei Cloud platform is roughly divided into five parts, namely: logging in to the IoTDA console, creating products, defining product models, registering devices, and generating device docking information[9].The platform-side development process is shown in Figure 7.

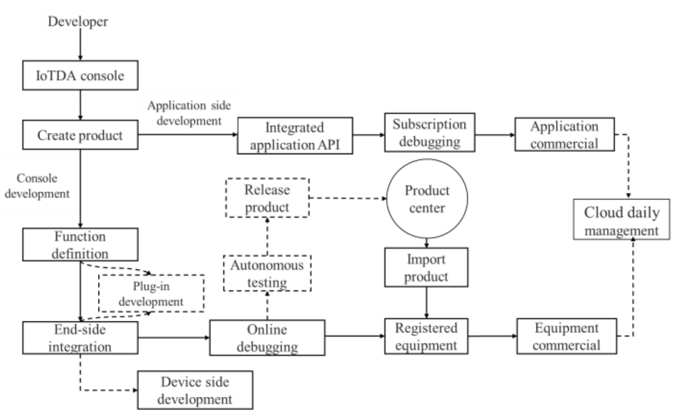

Figure 7. Cloud Platform Development Process

Create a product model on the Huawei Cloud Platform, including service IDs, attributes, commands, and so on. The product model is used to describe the capabilities and characteristics of the device.By defining the product model, developers can build an abstract model of a device on the Huawei Cloud Platform, enabling the platform to understand the services, attributes, commands, and other information supported by the device.The product model created on the Huawei Cloud Platform needs to be one-to-one mapped to the Hi3861 program, so that data reporting and command issuing can be completed. The product model is shown in Table 1.

Table 1. Product model definition list

Function name	identifier	data type	data definition
air temperature	Temperature	Int	Range: 0~100
air temperature	Humidity	Int	Range: 0~100
air humidity	soilmoisture	Int	Range: 0~100
Illumination intensity	Luminance	Int	Range: 0~100
wind motor	MotorStatus	String	ON,OFF
water pump	PumpStatus	String	ON,OFF

V. System testing

After the design of the system's hardware and software is completed, the hardware modules are assembled and integrated in a laboratory environment, and then the system is powered on. Since Wi-Fi data that needs to be connected was added during the programming process, the system can be connected to the set Wi-Fi after it is powered on, and then it can be connected to Huawei Cloud.The testing environment of this system is indoor. First, the soil moisture sensor is inserted into a flowerpot with a certain humidity, and the pump motor is placed in a bottle filled with water as the water source supply. The irrigation is achieved by connecting the hose to the pump, and the fan and other sensors are installed on the Hi3861 development board. The overall physical diagram of this system is shown in Figure 8.

Figure 8. Smart Agriculture System

By logging in to the Huawei Cloud Platform, it is possible to observe the environmental temperature, light intensity, soil humidity, and the status of water pumps and fans for crops. The data monitoring of the cloud platform is shown in Figure 9.The figure shows that the current environmental temperature is 25°C, the environmental humidity is 50RH, the light intensity is 25nit, the soil humidity is 64RH, and the status of the water pump and fan are both OFF. At this time, the environmental temperature measured by professional equipment is 25°C, the environmental humidity is 51RH, the light intensity is 25nit, and the soil humidity is 66RH, and the status of the water pump and fan are both OFF. It can be concluded that the data measured by this smart agricultural system is more in line with the actual environmental parameters, and the error is within the normal range.

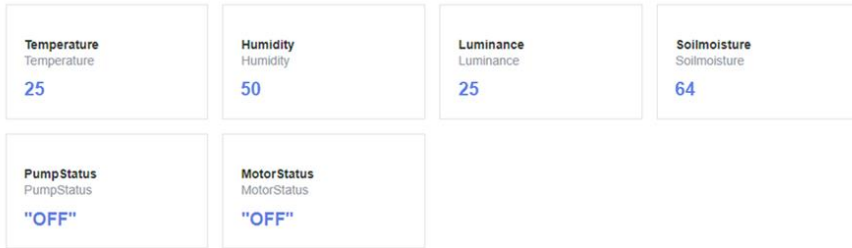

Figure 9. Cloud platform data monitoring diagram

By logging in to the Huawei Cloud platform, users can view the growth environment status of smart agricultural crops. Using the data storage function of the Huawei Cloud IoT platform, historical data related to environmental parameters in the physical model are displayed in the form of a line chart, which facilitates the viewing of historical data of the device, and up to 90 days of historical records can be viewed. The line chart of historical data of crop environment is shown in Figure 10.The figure shows the test environment monitoring sensitivity, and the human-induced environmental variables, such as the sudden increase of environmental temperature, humidity, and light intensity, to obtain the line chart shown in the figure[10]. The environmental temperature is 23 degrees, the humidity fluctuates between 55 and 58, and the light intensity can reach 75nit when there is external light source. After removing the light source, it drops to about 24nit, which verifies that the recorded historical data basically matches the actual situation.

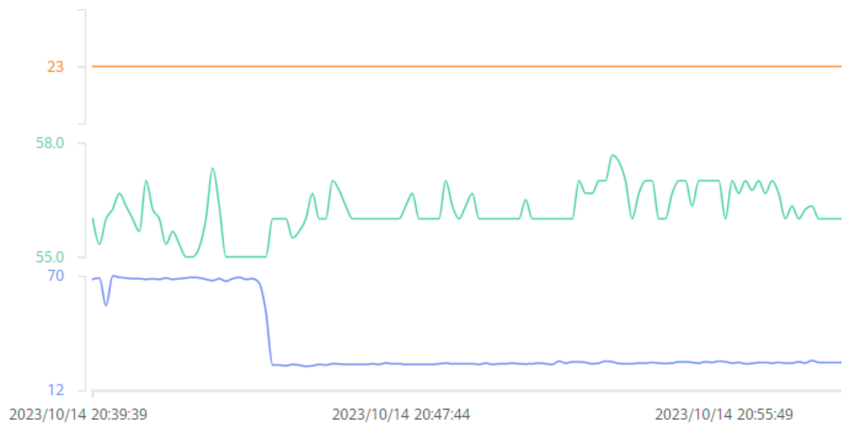

Figure 10. Line graph of historical data of crop environment

VI. Conclusions

This article uses Hi3861 as the main control chip, and the 2.4GHz Wi-Fi built into Hi3861 uses the MQTT protocol to connect to the Huawei cloud platform. Using this chip can easily, quickly, and cost-effectively implement device control and network connectivity functions. Managers can log in to the cloud platform through a PC web page or mobile client, and the cloud platform provides online real-time monitoring of crop growth environmental parameter changes and can store data in the cloud, providing a feasible practical solution for the development of smart agriculture.

References

[1] Sakthisudhan K ,N.Saravanakumar ,Mohanraj S . A Smart Agriculture System Based On Intel Galileo For Detection Of Cultivable Crops And Implemented On Soil Nutrients[J]. International Journal of Innovative Technology and Exploring Engineering (IJITEE),2019,8(9).

[2] A Smart Agriculture System Based On Intel Galileo For Detection Of Cultivable Crops And Implemented On Soil Nutrients[J]. International Journal of Innovative Technology and Exploring Engineering,2019,8(9).

[3] S. N ,K. A . Optimised fertiliser suggestion in smart agriculture system based on fuzzy inference rule[J]. Acta Agriculturae Scandinavica, Section B – Soil & Plant Science,2021,71(3).

[4] [Internet of Things Based on Smart Agriculture System Using Predictive Analysis[J]. International Journal of Recent Trends in Engineering and Research,2018,4(3).

[5] Galang M W S ,Erma S ,Agus R H . Increasing productivity of rice plants based on IoT (Internet Of Things) to realize Smart Agriculture using System Thinking approach[J]. Procedia Computer Science, 2022,197.

[6] Kannadhasan S ,Shanmuganantham M . Agriculture Monitoring and Smart Irrigation System Based on Wireless Sensors[J]. International Journal of Sensors and Sensor Networks,2019,7(4).

[7] Yile W ,Hanbing L ,Xian S B T , et al. Image Detection System Based on Smart Sensor Network and Ecological Economy in the Context of Fine Agriculture[J]. Journal of Sensors,2022,2022.

[8] Srinivas A ,Anna S ,V. R , et al. Agriculture Pump based on Smart Energy Harvesting on Solar System[J]. IOP Conference Series: Materials Science and Engineering,2020,981(4).

[9] Liu Y . Design and Implementation of Smart Agriculture System Based on Wireless Sensors Networks[J]. Advances in Computer, Signals and Systems,2023,7(5).

[10] Paola F ,Nicolas H ,Gonzalo T , et al. Eco-Efficient Systems Based on Nanocarriers for the Controlled Release of Fertilizers and Pesticides: Toward Smart Agriculture.[J]. Nanomaterials (Basel, Switzerland), 2023,13(13).